

Differential Equations:

An equation containing at least one derivative of a dependent variable with respect to an independent variable is called a differential equation.

For example: $y \frac{dy}{dx} + 2x = 0$, $x \frac{d^2y}{dx^2} + \frac{dy}{dx} - 2x = 0$ etc.

Order of Differential Equation:

The order of differential equation is the order of the highest derivative in the equation.

For example:

$y \frac{dy}{dx} + 2x = 0$ is first order differential equation.

$x \frac{d^2y}{dx^2} - 2x = 0$ is second order differential equation.

Solution of differential Equation:

A solution of differential equation is a relation between the variables (not involving derivatives) which satisfies the differential equation.

Initial value conditions:

The arbitrary constants involving in the solution of a differential equation can be determined by the given conditions. Such conditions are called initial value conditions.

Note:

- (i) A differential equation containing only derivative of first order can be written in terms of differentials.
- (ii) General solution of differential equation of order n contains n arbitrary constants which can be determined by n initial value conditions.

EXERCISE 3.8

Q.1 Check that each of the following equations written against the differential equation is its solution.

(i) $x \frac{dy}{dx} = 1 + y$, $y = cx - 1$

Solution:

$$y = cx - 1$$

Differentiate w.r.t. "x"

$$\frac{dy}{dx} = \frac{d}{dx}(cx - 1)$$

$$\frac{dy}{dx} = c$$

$$\text{Also } x \frac{dy}{dx} = 1 + y \dots (i)$$

$$\Rightarrow xc = 1 + cx - 1$$

$$cx = cx$$

Hence $y = cx - 1$ is the solution of (i)

Alternate solution:

$$x \frac{dy}{dx} = 1 + y$$

$$xdy = (1 + y)dx$$

$$\frac{1}{1+y} dy = \frac{1}{x} dx$$

Integrating both sides

$$\int \frac{1}{1+y} dy = \int \frac{1}{x} dx$$

$$\ln(1+y) = \ln x + \ln c$$

$$\ln(1+y) = \ln cx$$

$$1+y = cx$$

$$y = cx - 1$$

(ii) $x^2(2y+1) \frac{dy}{dx} - 1 = 0$, $y^2 + y = c - \frac{1}{x}$

Solution:

$$y^2 + y = c - \frac{1}{x}$$

Differentiate w.r.t. "x"

$$2y \frac{dy}{dx} + \frac{dy}{dx} = 0 - \left(\frac{-1}{x^2} \right)$$

$$(2y+1) \frac{dy}{dx} = \frac{1}{x^2}$$

$$\text{Also } x^2(2y+1) \frac{dy}{dx} - 1 = 0 \dots (i)$$

$$x^2 \left(\frac{1}{x^2} \right) - 1 = 0$$

$$1 - 1 = 0$$

Hence $y^2 + y = c - \frac{1}{x}$ is the solution

of (i)

$$(iii) \quad y \frac{dy}{dx} - e^{2x} = 1, \quad y^2 = e^{2x} + 2x + c$$

Solution:

$$y^2 = e^{2x} + 2x + c$$

Differentiate w.r.t. "x"

$$2y \frac{dy}{dx} = 2e^{2x} + 2$$

Dividing both sides by 2

$$y \frac{dy}{dx} = e^{2x} + 1$$

$$\text{Also } y \frac{dy}{dx} - e^{2x} = 1 \dots (i)$$

(i) becomes

$$e^{2x} + 1 - e^{2x} = 1$$

Hence $y^2 = e^{2x} + 2x + c$ is the solution of (i)

$$(iv) \quad \frac{1}{x} \frac{dy}{dx} - 2y = 0, \quad y = ce^{x^2}$$

Solution:

$$y = ce^{x^2}$$

Differentiate w.r.t. "x"

$$\frac{dy}{dx} = ce^{x^2} (2x)$$

$$\frac{dy}{dx} = 2cxe^{x^2}$$

$$\text{Also } \frac{1}{x} \frac{dy}{dx} - 2y = 0 \dots (i)$$

(i) becomes

$$\frac{1}{x} (2cxe^{x^2}) - 2(ce^{x^2}) = 0$$

$$2ce^{x^2} - 2ce^{x^2} = 0$$

Hence $y = ce^{x^2}$ is the solution of (i)

$$(v) \quad \frac{dy}{dx} = \frac{y^2 + 1}{e^{-x}}, \quad y = \tan(e^x + c)$$

Solution:

$$y = \tan(e^x + c)$$

Differentiate w.r.t. "x"

$$\frac{dy}{dx} = \sec^2(e^x + c) e^x$$

$$\text{Also } \frac{dy}{dx} = \frac{y^2 + 1}{e^{-x}} \dots (i)$$

(i) becomes

$$e^x \sec^2(e^x + c) = \frac{\tan^2(e^x + c) + 1}{e^{-x}}$$

$$e^x \sec^2(e^x + c) = e^x \sec^2(e^x + c)$$

Hence $y = \tan(e^x + c)$ is the solution of (i)

Solve the following differential equations:

$$Q.2 \quad \frac{dy}{dx} = -y$$

Solution:

$$\frac{dy}{dx} = -y$$

By separation of variables

$$\frac{dy}{y} = -dx$$

Integrating both sides

$$\int \frac{dy}{y} = -\int 1 dx$$

$$\ln y = -x + \ln c$$

$$\ln y = \ln e^{-x} + \ln c$$

$$\ln y = \ln ce^{-x}$$

$$y = ce^{-x}$$

$$Q.3 \quad ydx + xdy = 0$$

Solution:

$$ydx + xdy = 0$$

By separation of variables

$$xdy = -ydx$$

$$\frac{dy}{y} = \frac{-dx}{x}$$

Integrating both sides

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\ln y = -\ln x + \ln c$$

$$\ln x + \ln y = \ln c$$

$$\ln xy = \ln c$$

$$xy = c$$

$$Q.4 \quad \frac{dy}{dx} = \frac{1-x}{y}$$

Solution:

$$\frac{dy}{dx} = \frac{1-x}{y}$$

By separation of variables

$$y dy = (1-x) dx$$

Integrating both sides

$$\int y dy = \int (1-x) dx$$

$$\frac{y^2}{2} = x - \frac{x^2}{2} + c_1$$

$$y^2 = 2x - x^2 + 2c_1$$

$$y^2 = 2x - x^2 + c \quad \because 2c_1 = c$$

$$y^2 = x(2-x) + c$$

$$\text{Q.5} \quad \frac{dy}{dx} = \frac{y}{x^2}, (y > 0)$$

Solution:

$$\frac{dy}{dx} = \frac{y}{x^2}$$

By separation of variables

$$\frac{dy}{y} = \frac{dx}{x^2}$$

Integrating both sides

$$\int \frac{dy}{y} = \int x^{-2} dx$$

$$\ln y = \frac{x^{-1}}{-1} + \ln c$$

$$\ln y = \frac{-1}{x} + \ln c$$

$$\ln y = \ln e^{\frac{-1}{x}} + \ln c$$

$$\ln y = \ln ce^{\frac{-1}{x}}$$

$$y = ce^{\frac{-1}{x}}$$

$$\text{Q.6} \quad \sin y \operatorname{cosec} x \frac{dy}{dx} = 1$$

Solution:

$$\sin y \operatorname{cosec} x \frac{dy}{dx} = 1$$

By separation of variables

$$\sin y dy = \frac{dx}{\operatorname{cosec} x}$$

Integrating both sides

$$\int \sin y dy = \int \sin x dx$$

$$-\cos y = -\cos x + c_1$$

$$\cos y = \cos x + c \quad \because -c_1 = c$$

$$\text{Q.7} \quad x dy + y(x-1) dx = 0$$

Solution:

$$x dy + y(x-1) dx = 0$$

By separation of variables

$$x dy = -y(x-1) dx$$

$$\frac{dy}{y} = -\left(\frac{x-1}{x}\right) dx$$

$$\frac{dy}{y} = \left(\frac{-x}{x} + \frac{1}{x}\right) dx$$

$$\frac{dy}{y} = \left(-1 + \frac{1}{x}\right) dx$$

Integrating both sides

$$\int \frac{dy}{y} = \int -1 dx + \int \frac{1}{x} dx$$

$$\ln y = -x + \ln x + \ln c$$

$$\ln y - \ln x - \ln c = -x$$

$$\ln\left(\frac{y}{xc}\right) = -x$$

$$\frac{y}{xc} = e^{-x}$$

$$y = cxe^{-x}$$

$$\text{Q.8} \quad \frac{x^2+1}{y+1} = \frac{x}{y} \frac{dy}{dx}$$

Solution:

$$\frac{x^2+1}{y+1} = \frac{x}{y} \frac{dy}{dx}$$

By separation of variables

$$\frac{x}{y} \frac{dy}{dx} = \frac{x^2+1}{y+1}$$

$$\left(\frac{y+1}{y}\right) dy = \left(\frac{x^2+1}{x}\right) dx$$

$$\left(\frac{y}{y} + \frac{1}{y}\right) dy = \left(\frac{x^2}{x} + \frac{1}{x}\right) dx$$

$$\left(1 + \frac{1}{y}\right) dy = \left(x + \frac{1}{x}\right) dx$$

Integrating both sides

$$\int \left(1 + \frac{1}{y}\right) dy = \int \left(x + \frac{1}{x}\right) dx$$

$$y + \ln y = \frac{x^2}{2} + \ln x + \ln c$$

$$\ln y - \ln x - \ln c = \frac{x^2}{2} - y$$

$$\ln\left(\frac{y}{xc}\right) = \frac{x^2}{2} - y$$

$$\frac{y}{xc} = e^{\frac{x^2}{2} - y}$$

$$\frac{y}{xc} = e^{\frac{x^2}{2}} e^{-y}$$

$$\frac{y}{xc} = \frac{e^{\frac{x^2}{2}}}{e^y}$$

$$ye^y = cxe^{\frac{x^2}{2}}$$

Q.9 $\frac{1}{x} \frac{dy}{dx} = \frac{1}{2}(1+y^2)$

Solution:

$$\frac{1}{x} \frac{dy}{dx} = \frac{1}{2}(1+y^2)$$

$$\frac{dy}{1+y^2} = \frac{1}{2} x dx$$

Integrating both sides

$$\int \frac{dy}{1+y^2} = \frac{1}{2} \int x dx$$

$$\tan^{-1} y = \frac{1}{2} \frac{x^2}{2} + c$$

$$\tan^{-1} y = \frac{x^2}{4} + c$$

Q.10 $2x^2 y \frac{dy}{dx} = x^2 - 1$

Solution:

$$2x^2 y \frac{dy}{dx} = x^2 - 1$$

By separation of variables

$$2y dy = \frac{x^2 - 1}{x^2} dx$$

$$2y dy = \left(1 - \frac{1}{x^2}\right) dx$$

Integrating both sides

$$\int 2y dy = \int \left(1 - x^{-2}\right) dx$$

$$\frac{2y^2}{2} = x + \frac{1}{x} + c$$

$$y^2 = x + \frac{1}{x} + c$$

Q.11 $\frac{dy}{dx} + \frac{2xy}{2y+1} = x$

Solution:

$$\frac{dy}{dx} + \frac{2xy}{2y+1} = x$$

$$\begin{aligned} \frac{dy}{dx} &= x - \frac{2xy}{2y+1} \\ &= \frac{2xy + x - 2xy}{2y+1} \end{aligned}$$

$$\frac{dy}{dx} = \frac{x}{2y+1}$$

$$(2y+1)dy = x dx$$

Integrating both sides

$$\int (2y+1) dy = \int x dx$$

$$\frac{2y^2}{2} + y = \frac{x^2}{2} + c$$

$$y^2 + y = \frac{x^2}{2} + c$$

$$y(y+1) = \frac{x^2}{2} + c$$

Q.12 $(x^2 - yx^2) \frac{dy}{dx} + y^2 + xy^2 = 0$

Solution:

$$(x^2 - yx^2) \frac{dy}{dx} + y^2 + xy^2 = 0$$

By separation of variables

$$x^2(1-y) \frac{dy}{dx} + y^2(1+x) = 0$$

$$x^2(1-y) \frac{dy}{dx} = -y^2(1+x)$$

$$\left(\frac{1-y}{y^2}\right) dy = -\frac{(1+x)}{x^2} dx$$

$$\left(\frac{1}{y^2} - \frac{y}{y^2}\right) dy = -\left(\frac{1}{x^2} + \frac{x}{x^2}\right) dx$$

$$\left(\frac{1}{y^2} - \frac{1}{y}\right) dy = -\left(\frac{1}{x^2} + \frac{1}{x}\right) dx$$

Integrating both sides

$$\int \left(y^{-2} - \frac{1}{y}\right) dy = -\int \left(x^{-2} + \frac{1}{x}\right) dx$$

$$\frac{y^{-1}}{-1} - \ln y = -\frac{1}{(-x)} - \ln x - c$$

$$-\frac{1}{y} - \ln y = \frac{1}{x} - \ln x - c$$

$$\ln y + \frac{1}{y} = \ln x - \frac{1}{x} + c$$

Q.13 $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Solution:

$$\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$$

By separation of variables

$$\sec^2 y \tan x dy = -\sec^2 x \tan y dx$$

$$\frac{\sec^2 y dy}{\tan y} = -\frac{\sec^2 x}{\tan x} dx$$

Integrating both sides

$$\int \frac{\sec^2 y dy}{\tan y} = -\int \frac{\sec^2 x}{\tan x} dx$$

$$\ln \tan y = -\ln \tan x + \ln c$$

$$\ln \tan y + \ln \tan x = \ln c$$

$$\ln \tan x \tan y = \ln c$$

$$\tan x \tan y = c$$

Q.14 $y - x \frac{dy}{dx} = 2 \left(y^2 + \frac{dy}{dx} \right)$

Solution:

$$y - x \frac{dy}{dx} = 2 \left(y^2 + \frac{dy}{dx} \right)$$

By separation of variables

$$y - x \frac{dy}{dx} = 2y^2 + 2 \frac{dy}{dx}$$

$$-x \frac{dy}{dx} - 2 \frac{dy}{dx} = 2y^2 - y$$

$$-(x+2) \frac{dy}{dx} = 2y^2 - y$$

$$\frac{dy}{2y^2 - y} = \frac{-dx}{x+2}$$

$$\frac{dy}{y(2y-1)} = -\frac{dx}{x+2} \dots (i)$$

$$\text{Let } \frac{1}{y(2y-1)} = \frac{A}{y} + \frac{B}{2y-1} \dots (ii)$$

Multiplying both sides by $y(2y-1)$

$$1 = A(2y-1) + B(y) \dots (iii)$$

Put $y = 0$ in eq. (iii)

$$1 = A(-1) \Rightarrow \boxed{A = -1}$$

$$\text{Put } 2y-1=0 \Rightarrow y = \frac{1}{2} \text{ in eq. (iii)}$$

$$1 = A(0) + B\left(\frac{1}{2}\right) \Rightarrow \boxed{B = 2}$$

Put values of A and B in (ii)

$$\frac{1}{y(2y-1)} = \frac{-1}{y} + \frac{2}{2y-1}$$

(i) becomes

$$\left(\frac{-1}{y} + \frac{2}{2y-1} \right) dy = -\frac{dx}{x+2}$$

Integrating both sides

$$\int \left(\frac{-1}{y} + \frac{2}{2y-1} \right) dy = -\int \frac{dx}{x+2}$$

$$-\ln y + \ln(2y-1) = -\ln(x+2) - \ln c$$

$$\ln y - \ln(2y-1) = \ln(x+2) + \ln c$$

$$\ln \frac{y}{2y-1} = \ln(c(x+2))$$

$$\frac{y}{2y-1} = c(x+2)$$

Q.15 $1 + \cos x \tan y \frac{dy}{dx} = 0$

Solution:

$$1 + \cos x \tan y \frac{dy}{dx} = 0$$

By separation of variables

$$\cos x \tan y \frac{dy}{dx} = -1$$

$$\tan y dy = \frac{-dx}{\cos x}$$

Integrating both sides

$$\int \tan y dy = -\int \sec x dx$$

$$-\ln \cos y = -\ln(\sec x + \tan x) - \ln c$$

$$\ln \cos y = \ln(\sec x + \tan x) + \ln c$$

$$\ln \cos y = \ln(c(\sec x + \tan x))$$

$$\cos y = c(\sec x + \tan x)$$

Q.16 $y - x \frac{dy}{dx} = 3 \left(1 + x \frac{dy}{dx} \right)$

Solution:

$$y - x \frac{dy}{dx} = 3 \left(1 + x \frac{dy}{dx} \right)$$

By separation of variables

$$y - x \frac{dy}{dx} = 3 + 3x \frac{dy}{dx}$$

$$y-3=4x\frac{dy}{dx}$$

$$4x\frac{dy}{dx}=y-3$$

$$\frac{dy}{y-3}=\frac{dx}{4x}$$

Integrating both sides

$$\int \frac{dy}{y-3} = \frac{1}{4} \int \frac{dx}{x}$$

$$\ln(y-3) = \frac{1}{4} \ln x + \ln c$$

$$\ln(y-3) = \ln x^{\frac{1}{4}} + \ln c$$

$$\ln(y-3) = \ln\left(cx^{\frac{1}{4}}\right)$$

$$y-3 = cx^{\frac{1}{4}}$$

$$y = 3 + cx^{\frac{1}{4}}$$

Q.17 $\sec x + \tan y \frac{dy}{dx} = 0$

Solution:

$$\sec x + \tan y \frac{dy}{dx} = 0$$

By separation of variables

$$\tan y \frac{dy}{dx} = -\sec x$$

$$\tan y dy = -\sec x dx$$

Integrating both sides

$$\int \tan y dy = -\int \sec x dx$$

$$-\ln \cos y = -\ln(\sec x + \tan x) - \ln c$$

$$\ln \cos y = \ln(\sec x + \tan x) + \ln c$$

$$\ln(\cos y) = \ln(c(\sec x + \tan x))$$

$$\cos y = c(\sec x + \tan x)$$

Q.18 $(e^x + e^{-x}) \frac{dy}{dx} = e^x - e^{-x}$

Solution:

$$(e^x + e^{-x}) \frac{dy}{dx} = e^x - e^{-x}$$

$$\frac{dy}{dx} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$dy = \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

Integrating both sides

$$\int dy = \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

$$\because \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

$$y = \ln|e^x + e^{-x}| + c$$

Q.19 Find the general solution of the equation $\frac{dy}{dx} - x = xy^2$. Also find the particular solution if $y=1$ when $x=0$

Solution:

$$\frac{dy}{dx} - x = xy^2$$

By separation of variables

$$\frac{dy}{dx} = x + xy^2$$

$$\frac{dy}{dx} = x(1 + y^2)$$

$$\frac{dy}{1 + y^2} = x dx$$

Integrating both sides

$$\int \frac{dy}{1 + y^2} = \int x dx$$

$$\tan^{-1} y = \frac{x^2}{2} + c \dots (i)$$

is the general solution

put $y=1, x=0$ in (i)

$$\tan^{-1}(1) = c$$

$$\frac{\pi}{4} = c$$

Put value of c in (i)

$$\tan^{-1} y = \frac{x^2}{2} + \frac{\pi}{4}$$

Q.20 Solve the differential equation

$$\frac{dx}{dt} = 2x, \text{ given that } x=4 \text{ when } t=0.$$

Solution:

$$\frac{dx}{dt} = 2x$$

By separation of variables

$$\frac{dx}{x} = 2dt$$

Integrating both sides

$$\int \frac{dx}{x} = 2 \int dt$$

$$\ln x = 2t + \ln c$$

$$\ln x = \ln e^{2t} + \ln c$$

$$\ln x = \ln ce^{2t}$$

$$x = ce^{2t} \dots \text{(i) is the general solution}$$

$$\text{Put } x = 4, \quad t = 0 \text{ in (i)}$$

$$\ln 4 = c$$

$$\text{Put value of } c \text{ in (i)}$$

$$\ln x = 2t + \ln 4$$

$$\ln x - \ln 4 = 2t$$

$$\ln\left(\frac{x}{4}\right) = 2t$$

$$\frac{x}{4} = e^{2t}$$

$$x = 4e^{2t}$$

Q.21 Solve the differential equation

$$\frac{ds}{dt} + 2st = 0. \quad \text{Also find the}$$

particular solution if } s = 4e,

when } t = 0.

Solution:

$$\frac{ds}{dt} + 2st = 0$$

By separation of variables

$$\frac{ds}{dt} = -2st$$

$$\frac{ds}{s} = -2tdt$$

Integrating both sides

$$\int \frac{ds}{s} = -2 \int t dt$$

$$\ln s = -\frac{2t^2}{2} + \ln c$$

$$\ln s = \ln e^{-t^2} + \ln c$$

$$\ln s = \ln ce^{-t^2}$$

$$s = ce^{-t^2} \dots \text{(i)}$$

is the general solution

$$\text{Put } s = 4e, t = 0 \text{ in (i)}$$

$$4e = c$$

Put value of } c \text{ in eq. (i)}

$$s = 4ee^{-t^2}$$

$$s = 4e^{1-t^2}$$

Q.22 In a culture, bacteria increases at

the rate proportional to the number of bacteria present. If bacteria are 200 initially and are doubled in 2 hours, find the number of bacteria present four hours later.

Solution:

Let } p \text{ be the number of bacteria present, then}

$$\frac{dp}{dt} \propto p$$

$$\frac{dp}{dt} = kp$$

By separation of variables

$$\frac{dp}{p} = kdt$$

Integrating both sides

$$\int \frac{dp}{p} = \int kdt$$

$$\ln p = kt + \ln c$$

$$= \ln e^{kt} + \ln c$$

$$\ln p = \ln ce^{kt}$$

$$p = ce^{kt} \dots \text{(i)}$$

$$\text{Put } p = 200, t = 0 \text{ in (i)}$$

$$200 = c$$

Hence (i) becomes

$$p = 200e^{kt} \dots \text{(ii)}$$

$$\text{Put } t = 2, p = 400 \text{ in (ii)}$$

$$400 = 200 e^{2k}$$

$$2 = e^{2k}$$

$$\ln 2 = 2k \Rightarrow k = \frac{1}{2} \ln 2$$

Hence

$$p = 200e^{\left(\frac{1}{2} \ln 2\right)t}$$

If } t = 4, \text{ then}

$$p = 200e^{\left(\frac{1}{2} \ln 2\right)(4)}$$

$$p = 200e^{2 \ln 2}$$

$$p = 200e^{\ln 4}$$

$$p = 200(4)$$

$$p = 800$$

which is required number of bacteria.

Q.23 A ball is thrown vertically upward with a velocity of 2450 cm/sec.

- Neglecting air resistance, find**
(i) Velocity of ball at any time 't'
(ii) Distance traveled in any time 't'
(iii) Maximum height attained by the ball.

Solution:

Let v be the velocity of ball at any time t , then

(i) $\frac{dv}{dt} = -g$

By separation of variables

$$dv = -gdt$$

Integrating both sides

$$\int dv = -g \int dt$$

$$v = -gt + c_1 \dots (i)$$

put $v = 2450$, $t = 0$ in (i)

$$2450 = c_1$$

(i) becomes

$$v = -gt + 2450$$

$$v = 2450 - 980t \dots (ii)$$

Where $g = 980 \text{ cm/sec}^2$

- (ii)** Let ' h ' be the height of ball at any time t , then

$$\frac{dh}{dt} = 2450 - 980t$$

By separation of variables

$$dh = (2450 - 980t) dt$$

Integrating both sides

$$\int dh = \int (2450 - 980t) dt$$

$$h = 2450t - \frac{980t^2}{2} + c_2$$

$$h = 2450t - 490t^2 + c_2 \dots (iii)$$

put $h = 0$, $t = 0$ in (iii)

$$0 = 0 - 0 + c_2 \Rightarrow c_2 = 0$$

Then (iii) becomes

$$h = 2450t - 490t^2 \dots (iv)$$

- (iii)** Maximum height is attained by the ball when $v = 0$

$$\text{i.e. } 2450 - 980t = 0$$

$$2450 = 980t \Rightarrow t = \frac{2450}{980}$$

$$t = 2.5 \text{ sec}$$

From (iv) the maximum height attained in cm is

$$= 2450 \times (2.5) - 490(2.5)^2$$

$$= 6125 - 3062.5$$

$$= 3062.5 \text{ cm}$$